

Kepler

The overreach, and what survived it

Two models of the solar system, published by the same man, twenty-three years apart.
One of them is still in use. You are going to find out which — by checking.



WHO WE ARE TALKING ABOUT

A man who wanted the universe to be beautiful



1571-1630

Württemberg, Graz, Prague, Linz. Lutheran in a Catholic empire, repeatedly expelled.



The day job

Imperial Mathematician — which mostly meant casting horoscopes for the Emperor.



The data

Tycho Brahe's observations. The most precise in Europe, and Kepler spent years fighting the family for access to them.



The temperament

He believed God was a geometer, and that the solar system had to be built on shapes.

The five solids

There are exactly five regular solids — cube, tetrahedron, octahedron, dodecahedron, icosahedron. Euclid proved the list complete. Kepler knew six planets, which leaves five gaps between them.

Five solids. Five gaps. He thought that could not be a coincidence.

- Nest each solid between two planetary spheres
- Each solid fixes a ratio of outer radius to inner radius
- That ratio is pure geometry — no measurement, no fitting

THE NESTING, OUTWARD

octahedron

Mercury → Venus

icosahedron

Venus → Earth

dodecahedron

Earth → Mars

tetrahedron

Mars → Jupiter

cube

Jupiter → Saturn

The third law

$$T^2 \propto a^3$$

period squared goes as distance cubed

Twenty-three years later, in a book mostly about musical harmony, Kepler stated a relationship between how long a planet takes to go round and how far out it sits.

He gave it four sentences.

He thought the solids were the important part.

WHAT MAKES THIS DIFFERENT

- It is not a shape. It is a proportion — it says nothing about why.
- It was extracted from Tycho's measurements, not derived from geometry.
- Kepler considered it decidedly the lesser of his two results.

Both models were built from the same six planets and the same observer's data.

Two columns, ten minutes

COLUMN 1

$$T^2 \div a^3$$

For each of six planets. If the third law holds, this column does something recognisable. Six rows.

COLUMN 2

predicted $\div$ actual

The five solid ratios are given to you. Compare each against the real spacing. Five rows.

Everything you need is in the spreadsheet. No algebra, no derivations — Tycho's numbers are already typed in, and the solid ratios are already computed. You are doing division and comparing sizes.

Post your prediction

Two sentences. Which model fits better, and roughly how much better?

Graded on

Being posted on time, and being specific enough to be wrong.

Not graded on

Being correct. A confident wrong answer earns full marks.

Why it is locked

You cannot read anyone else's post until you have written yours.

Your tutor

Kepler is in the tutor deck. In "before" mode he will not give you the answer — he does not have it.

End of Part One. Part Two unlocks after the computation deadline.

What the numbers did

You have the two columns in front of you. Let us read them together.



The fitted one held

Planet	$T^2 \div a^3$	off by
Mercury	0.9996	0.04 %
Venus	1.0002	0.02 %
Earth	1.0000	0.00 %
Mars	1.0000	0.00 %
Jupiter	0.9985	0.15 %
Saturn	0.9861	1.39 %

1.4 %

worst case, across the whole set

Saturn sits forty times further out than Mercury. The ratio holds across that entire range, and the one visible drift — Saturn — is the planet Tycho had watched for the shortest fraction of an orbit.

VERIFIED

arithmetic, reproducible from the sheet

The derived one did not

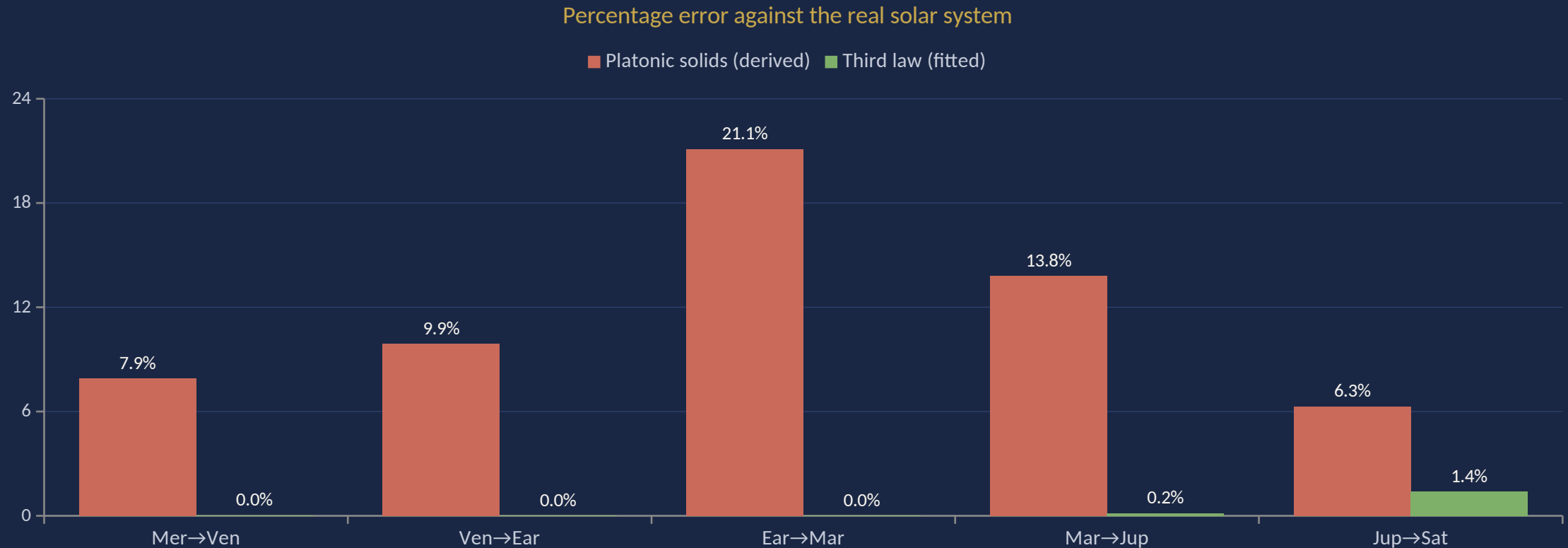
Gap	Solid	predicted	actual	off by
Mercury → Venus	octahedron	1.732	1.869	+7.9 %
Venus → Earth	icosahedron	1.258	1.383	+9.9 %
Earth → Mars	dodecahedron	1.258	1.524	+21.1 %
Mars → Jupiter	tetrahedron	3.000	3.415	+13.8 %
Jupiter → Saturn	cube	1.732	1.841	+6.3 %

6 – 21 %

every gap, all five wrong in the same direction

Not one of the five is close. And they all miss the same way — the real planets sit further apart than the solids allow.

Same man. Same data. Same decade.



What this establishes — and what it does not

What the computation shows

The five solid ratios do not match the real spacings. All five miss, all in the same direction.

$T^2 \div a^3$ is constant across six planets to within 1.4%.

What it does NOT show

That the third law is true. Six planets is six data points — it is consistent with them, which is weaker.

That deriving from mathematics is worse than fitting. That is not the lesson.

The real difference is what happened next. The third law was later checked against planets Kepler never saw, against the moons of Jupiter, and against exoplanets found four centuries after his death. It kept holding. The solids were never checked against anything else, because there was nothing else to check them against — five gaps was all the model ever had.

A model that can only be checked
against the data that made it
has not been checked.

This week's reading

Kepler's Other Correspondents · [book7/ch-keplers-correspondents.html](#) — written above this level on purpose. Use the Kepler tutor to get through it, and remember the chapter outranks the tutor.